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Overview

Apply the derisking toolkit to identify, prioritize, and **test** your venture's most critical assumptions using **primary behavioral data** and well-designed experiments (prototypes, pilots, or equivalent experiments). You will present a **complete derisking plan** and evidence from executed experiments in your coach meeting.

Part A — Derisking Plan (Five Steps)

1. **Decompose (20+ hypotheses):** List granular unknowns across demand, supply/tech, value, operations, partners, unit economics, etc. Break big risks into smaller, testable pieces.

Hypothesis Risks:

Demand & Customer Value Hypotheses

1. Venues experience enough safety-related pain (liability, crowd issues, staffing gaps) that they are willing to pay for an AI-powered safety system.
2. Venue managers perceive measurable ROI from reduced risk, insurance benefits, or improved incident response times.
3. Customers trust AI-based safety monitoring more than (or equal to) human-only monitoring.
4. Pilot venues will convert to paying customers after initial trials if early incidents are prevented/flagged.
5. Event hosts believe the benefits outweigh the cost, even though the product does not directly increase revenue.

Technology & Product Hypotheses

6. Models can reliably detect stress, suspicious behavior, weapons, and anomalies at >X accuracy thresholds in real-world conditions.
7. Decision-making thresholds for alerts can be tuned to reduce false positives to an acceptable level.
8. The system can integrate with a wide variety of existing camera infrastructure (different brands, resolutions, layouts).

9. Camera placement + our recommended hardware is sufficient to eliminate blind spots in high-density areas.
10. The system can operate in real-time with low latency across diverse network environments.
11. The system remains failsafe during critical live events (overload, packet loss, sensor downtime).
12. Our models can be continuously retrained with new data without violating privacy or losing performance.

Compliance, Safety, Privacy Hypotheses

13. AI monitoring should remain compliant with state and federal privacy laws (NYC, state-specific biometric laws, etc.).
14. The solution can stay updated with evolving public-space capacity and safety regulations.
15. Training data pipelines can meet strict data privacy standards (PII removal, secure storage).
16. Insurance companies recognize this technology as risk-reducing, contributing to potential cost savings for buyers.

Operational & GTM (Go To Market) Hypotheses

17. We can source the engineering talent required to build the ML models and platform.
18. We can deliver rapid onboarding + staff training that avoids operational friction for venues.
19. Maintenance (hardware + software) can be done predictably and affordably for both small and large venues.
20. The product can scale nationally after initial NYC pilots without excessive custom integrations.

Partnerships, Supply, Unit Economics Hypotheses

21. Security camera manufacturers, insurers, and venue-management companies are willing to partner, reducing CAC and accelerating trust.
22. Unit economics remain positive after accounting for compute costs, installation, support, and hardware bundles.

2. **Prioritize (Risk Factor):** Score each hypothesis for *uncertainty* (1–3) and *importance* (1–3). Compute *Risk Factor = uncertainty × importance*.

Hypothesis	Uncertainty (1-3)	Importance (1-3)	Risk Factor (uncertainty x importance)
Venues experience enough safety-related pain (liability, crowd issues, staffing gaps) that they are willing to pay for an AI-powered safety system.	3	3	9
Venue managers perceive measurable ROI from reduced risk, insurance benefits, or improved incident response times.	2	3	6
Customers trust AI-based safety monitoring more than (or equal to) human-only monitoring.	1	1	1
Pilot venues will convert to paying customers after initial trials if early incidents are prevented/flagged.	2	2	4
Event hosts believe the benefits outweigh the cost, even though the product does not directly increase revenue.	2	1	2
Models can reliably detect stress, suspicious behavior, weapons, and anomalies at >X accuracy thresholds in real-world conditions.	2	3	6
Decision-making thresholds for alerts can be tuned to reduce false positives to an acceptable level.	2	3	6

The system can integrate with a wide variety of existing camera infrastructure (different brands, resolutions, layouts).	2	3	6
Camera placement + our recommended hardware is sufficient to eliminate blind spots in high-density areas.	1	3	3
The system can operate in real-time with low latency across diverse network environments.	2	3	6
The system remains failsafe during critical live events (overload, packet loss, sensor downtime).	2	3	6
Our models can be continuously retrained with new data without violating privacy or losing performance.	1	3	3
AI monitoring should remain compliant with state and federal privacy laws (NYC, state-specific biometric laws, etc.).	1	3	3
The solution can stay updated with evolving public-space capacity and safety regulations.	2	3	6
Training data pipelines can meet strict data privacy standards (PII removal, secure storage).	2	3	6
Insurance companies recognize this technology as risk-reducing,	2	3	6

contributing to potential cost savings for buyers.			
We can source the engineering talent required to build the ML models and platform.	1	3	3
We can deliver rapid onboarding + staff training that avoids operational friction for venues.	1	3	3
Maintenance (hardware + software) can be done predictably and affordably for both small and large venues.	1	3	3
The product can scale nationally after initial NYC pilots without excessive custom integrations.	3	1	3
Security camera manufacturers, insurers, and venue-management companies are willing to partner, reducing CAC and accelerating trust.	2	2	4
Unit economics remain positive after accounting for compute costs, installation, support, and hardware bundles.	1	2	2

3. **Map tasks & costs:** For each high-risk item, define a mitigation task and whether it uses *primary* (preferred) or *secondary* data; estimate time and cost.

ID	High-Risk Hypothesis (short)	Mitigation Task (what we'll do)	Data	Time	Cost ($)
H1 + H2	Venues faces enough safety related pain & Managers see measurable ROI	Atleast 3 interviews (venue GMs) + 1 underwriter; show mock "Risk Score → premium delta" and ask WTP	Primary	1-2 wk	0–100
H6	Models detect threats at target accuracy	Run YOLOv8/DeepSort on small clip set (public + staged); label vs. human baseline; compute P/R	Primary	2 wks	~200
H7	Thresholds keep false positives low	Sweep sensitivity on same clips; log FP/hr & miss rate; pick ops threshold	Primary	1 wk	~100
H8	Works with existing cams (multi-brand)	Ingest 3 sources (IP cam, low-res, phone RTSP); record setup time & frame stability	Primary	1 wk	~50
H10	Real-time, low latency	Edge test on Jetson/RPi; measure camera→alert E2E latency under load	Primary	1 wk	~150
H11	Failsafe under live-event stress	Simulate packet loss/overload; verify graceful degradation & recovery logs	Secondary	1 wk	~50
H14	Stays compliant w/ evolving regs	Desk review (NYC/NY biometric & signage); 30-min consult; produce 1-page compliance checklist	Secondary	1 wk	~200
H15	Privacy-safe training pipeline	Draft data-flow (PII removal, retention, access); peer review with privacy mentor	Secondary	1 wk	0
H16	Insurers recognize risk reduction	2 underwriter convos; react to sample "venue risk report"; note discount mechanisms	Primary	1 wk	~100
H12/ H20*	Continual retrain & scale nationally	Tiny MLOps dry-run: versioned model, swap test; document integration steps	Primary	1 wk	~100

*H12 (“retrain without privacy loss”) and H20 (“scale nationally w/o heavy custom work”) are tightly linked

Total est. cost: ≈ $850–900

Parallelizable: H2/H16; H6/H7; H8/H10/H11; H14/H15.

4. **Sequence:** Note dependencies or constraints. Plan for *maximum information return* early; fail fast, fail cheap.

A. Dependency Map (what must happen before what)

1. Market & Insurance Signal (A)
    - Covers: H2 ROI; H16 insurer recognition
    - Task(s): Interviews with venue GMs (show ROI mock) + 1–2 underwriters (react to “risk score → premium delta”).
    - Unlocks: Go/No-Go for pilot outreach and pricing.
2. Offline Tech Baselines (B)
    - Covers: H6 detection accuracy; H7 thresholding
    - Depends on: none
    - Unlocks: Confidence to approach venues with credible metrics; input to alert policies.
3. Infrastructure Compatibility ©
    - Covers: H8 multi-camera integration; H10 latency
    - Depends on: (B) model/alert pipeline running
    - Unlocks: Whether a pilot can run on existing hardware; edge vs cloud choice.
4. Resilience & Ops (D)
    - Covers: H11 failsafe under load
    - Depends on: (C) ingest path + alerting in place
    - Unlocks: Incident-response runbooks; SLO/SLA draft for pilot.
5. Compliance & Privacy (E)
    - Covers: H14 evolving regs; H15 privacy-safe training
    - Depends on: none (can run in parallel)
    - Unlocks: Signage language, data-handling policy, and allowable configurations for live tests.

B. One-Page Sequence (maximize early info, low cost)

Phase	Run	What you do	Depends on	Primary Risks
Week 1	**A** Market, Pain-point & Insurance Signal	3 venue GM interviews; 1–2 underwriter reactions to sample “risk report”	—	H1, H2, H16
	B1 Offline Accuracy Baseline	Label 50–100 short clips; measure P/R on threats; quick threshold sweep	—	H6, H7
Week 2	**B2** Threshold Policy	Pick operating point (FP/hr & miss rate); draft alert policy	B1	H7
	E1 Compliance Scan	Desk review NYC/NY biometrics & signage; 30-min consult; 1-page checklist	—	H14, H15
Week 3	**C** Ingest + Latency Smoke Test	Connect 2–3 camera types; measure E2E latency on edge device	B1	H8, H10
	E2 Privacy Dataflow	Finalize data-minimization & retention diagram; peer review	E1	H15
Week 4	**D** Resilience Simulation	Throttle network; drop frames; observe recovery; update runbook	C	H11
	Gate Pilot Readiness Review	Check pass/fail thresholds (see below)	A,B,C,E	H2,H6,H7, H8,H10,H14,H15,H16

C. Constraints to respect (and how we handle them)

- Access to feeds: Some venues won't share live streams. → Use public/staged clips for B1/B2; ask for short on-site tests only after Gate passes (keeps cost/ethics tight).
- Privacy/biometrics rules: NYC signage & PII concerns. → No facial ID storage, event-only buffered processing, and signage template from E1 (pre-approved).
- Hardware variability: Mixed camera brands/bitrates. → C tests multiple sources and documents minimum spec before any pilot.
- Compute cost creep: Guardrails for clip counts and log retention; log metrics not media whenever possible.
- False-positive risk: B2 sets FP/hr ceiling; alerts start as advisory (human-in-loop) during any pilot.

D. Fail-Fast gates (cheap, objective thresholds)

- Market gate (A): ≥2 venues express pilot interest and one underwriter confirms your risk report would inform pricing/discount discussion → else pivot messaging/pricing (no pilot outreach).
- Accuracy gate (B1): ≥0.80 precision and ≥0.75 recall on labeled clip set at selected operating point → else narrow scope to fewer classes or improve dataset.
- Latency gate (C): E2E (frame→alert) ≤1.0s at 720p/10fps on edge device → else optimize pipeline or lower resolution.
- Compliance gate (E1/E2): Checklist green-light on signage, retention, and no biometric storage → else re-design data path before any live test.

E. What runs in parallel (to compress timeline)

- A (market) ↔ B1/B2 (offline tech) ↔ E1 (compliance) all parallel in Weeks 1–2.
- C starts as soon as B1 yields a runnable model.
- D follows C; E2 can finalize while C runs.

F. Why this way (max info early approach)

- Week 1–2 produce the core signal: will buyers care? (A) and can it work at all? (B/E). Both are low-cost and decisive.
- Week 3–4 validate feasibility to pilot (C/D) only after there's buyer and compliance signal—avoids premature engineering.
- This ordering directly reflects Step-4 guidance to optimize information return and "fail fast, fail cheap."

5. **Select risk-reducing path:** Order tasks by (Risk Factor ÷ Cost), then minimally adjust for feasibility. This is your week-by-week plan.

A) Risk-Reducing Ordering (Risk Factor ÷ Cost)

[RF = Uncertainty × Importance from your table (items with RF=6 are the focus)]

ID	Hypothesis (short)	RF	Est. Cost ($)	RF/Cost	Task (shorthand)
H15	Privacy-safe training pipeline	6	**0**	→∞	Draft data-flow (no PII storage); peer review
H1+ H2	Market Adaptability & WTP	6	50	**0.12**	Interview venues and managers to understand Pain-points and WTP
H8	Works with existing cams	6	50	**0.12**	Ingest 3 camera sources; record setup/latency
H11	Failsafe under stress	6	50	**0.12**	Simulate loss/overload; observe recovery
H2	Managers see ROI	6	100	**0.06**	3 venue GM interviews + ROI mock review
H7	Thresholds keep FP low	6	100	**0.06**	Sensitivity sweep; pick ops threshold
H16	Insurers recognize risk reduction	6	100	**0.06**	2 underwriter reactions to “risk report”
H10	Real-time, low latency	6	150	**0.04**	Edge E2E latency test (frame→alert)
H6	Detect threats at target accuracy	6	200	**0.03**	Offline baseline on labeled clips
H14	Stay compliant w/ regs	6	200	**0.03**	Desk review + 1-pager checklist

Minimal feasibility adjustments (from Step 4):

- H6 before H7 (need a model to tune thresholds).
- H6 before H8/H10 (pipeline must exist before multi-cam ingest/latency).
- H8 before H11 (failsafe needs ingest path).
- These adjustments follow your Step-4 dependency map and still keep high RF/$ tasks early

B) Week-by-Week Plan (after dependency adjustments)

1. Week 1 — Cheap, parallel, decision-making signals
    - H1 & H2 Manager ROI interviews (primary)
    - H16 Underwriter reactions to risk report (primary)
    - H6 Offline accuracy baseline (primary; start labeling + first metrics)
    - H15 Privacy-safe pipeline diagram + peer review (secondary; $0)
    - Gate: If no venue interest and no insurer signal → pause pilot outreach.
2. Week 2 — Turn metrics into operations + compliance
    - H7 Threshold policy on Week-1 model (primary)
    - H14 Compliance desk review + signage/retention checklist (secondary)
    - Gate: Accuracy/thresholds acceptable (e.g., ≥0.80 precision & ≥0.75 recall) and compliance green-light → proceed to ingest tests.
3. Week 3 — Feasibility to run for real
    - H8 Multi-camera ingest (primary; 3 sources)
    - H10 Edge E2E latency measurement (primary)
    - Gate: Latency ≤1.0s at 720p/10fps and stable ingest on ≥2 camera types → proceed to resilience.
4. Week 4 — Resilience + pilot readiness
    - H11 Failsafe simulation under packet loss/overload (secondary/sim)
    - Pilot Readiness Review: Summarize Week 1–4 results; confirm venue(s) for a small, supervised pilot.
    - Outcome: If all gates pass, lock a tiny, low-cost pilot; else, iterate the failing link first (e.g., narrow detection classes or adjust data path).

C) Why this is the risk-reducing path

- Uses the assignment's RF/Cost ordering to maximize information per dollar, then only reorders where dependencies demand it
- Concentrates spend where RF/Cost is highest (H15, H8, H11, H2/H7/H16) while ensuring technical feasibility gates (H6→H7; H6→H8/H10; H8→H11) are met.
- Aligns with "fail fast, fail cheap" by front-loading interviews and offline tests before committing to any live pilot

Part B — Experiments (Design & Execution)

Experiment 1 (Primary) - Pain Point analysis and Willingness to pay

- **Summary:** The product's success hinges on solving a common venue security problem and addressing the shortcomings of current systems. While the ROI isn't directly measurable, a comprehensive risk assessment helps venue operators understand their vulnerabilities and the true cost of inaction. This understanding drives their willingness to invest in a solution that builds trust and mitigates critical risks.
- **Hypothesis** (falsifiable):
  - ≥50% of small venues find current CCTV + staff monitoring insufficient.
  - ≥20% of venues are willing to pay **≥$100/month**.
  - Subscription pricing converts ≥1.5× more pilots.
- **Method**
  - Participants: venue owners/managers across bars, halls, and restaurants.
  - What they do: short interview on security workflow, pain points, and early product concept.
  - Duration: 1 week; local outreach.
- **Variables & Instrumentation**
  - **Independent:** Pricing model, price shown, venue size.
  - **Dependent:** Stated WTP, likelihood-to-buy, pilot conversion, alert engagement.
  - **Measurement:** Survey logs, interview transcripts, pilot alert timestamps, Stripe payments.
- **Success Criteria**
  - ≥50% report significant pain (≥6/10)
  - ≥20% indicate WTP ≥$99/mo OR optimal price ≥$79.
  - Pilot conversion ≥15% and subscription model ≥1.5× better.
  - Pilot shows operational value (≥30% faster response or meaningful incident prevention).

Execution of the experiment 1

Pricing strategy

Starter Tier: $80/month ($960/ year) and one time setup $1500

- Target: Small bars, nightclubs, independent venues (100–250 capacity)
- Features: AI threat detection (up to 4-6 cameras), real-time alerts, monthly compliance reports, email support
- Customer Value: $12k–15k annual value, ~6x ROI, 2-3 month payback
- Year 1 Target: 40–50 venues, ~$100k–125k revenue

Professional Tier: $150/month ($1800/year) and one time $3000

- Target: Mid-sized arenas, stadiums, convention centers, theaters (250–500 capacity)
- Features: Unlimited cameras, advanced crowd anomaly detection, insurance integration, dedicated account manager
- Customer Value: $20k–25k annual value, ~5x ROI, 2–3 month payback
- Year 1 Target: 10–15 venues, ~$48k–75k revenue

Enterprise Tier: $1500/month ($18000/year) and one time setup $20k

- Target: Large stadiums, major venues (500–10,000+ capacity)
- Features: Multi-site aggregation, 24/7 managed SOC, white-label option, emergency response integration
- Customer Value: 100k+ annual value, ~3x ROI, 1–2 month payback
- Year 1 Target: 1–3 pilots/negotiations, ~$38k-115k revenue

Interviews

Summary: The team interviewed five venues ranging from small pubs and eateries to rooftop restaurants. The objective was to understand the key pain points in their current security systems and assess their willingness to pay for an automated security solution designed to enhance safety and strengthen audience trust

We presented a prototype demo and provided interviewee with a first-hand look at the concept.

Interview 1 - Granny Annies

Venue: Fine Dining Restaurant

Interviewees: Waiter, Manager

Pain Points

Primary: Monitoring non-permitted activities during events (smoking in non-smoking zones, littering)

- Bi-weekly trivia nights & monthly parties
- Manual staff enforcement ($50–75/month)
- Impacts staff efficiency & guest experience

Secondary: Front-door checking & entry management

- Manual verification, capacity checks, dress codes
- Cost: $15–25/month

Tertiary: Crowd management during events

- ~24 annual events strain operations
- Cost: $100–150/month (includes temp staff)

Low Priority: Security/threat detection

- Rarely faces incidents (fights, brawls)
- Fine-dining clientele = low-risk
- Current security already “fine”

Willingness to Pay

WTP Range: $75–125/month ($1,200/year)

Drivers:

- Trust retention (40%): Maintain brand reputation
- Compliance automation (35%): Reduce manual enforcement burden
- Operational efficiency (20%): Free staff for service
- Peace of mind (5%): Consistent enforcement

NOT Driven By:

- Threat prevention (rarely occurs)
- Growth/acquisition (limited event capacity, space-constrained)
- Hard ROI (WTP exceeds cost savings—this is a “peace of mind premium”)

Interview 2 - Jimmy’s Corner

Venue: Sports Dive Bar

Interviewee: Bartender/Owner

Pain Points

Primary: Bar brawls & aggressive behavior

- Frequent incidents (fights, conflicts among patrons)
- Manual security intervention required
- Liability & legal exposure from negligent security
- Reputation damage from violence incidents

Secondary: Limited staff capacity for crowd control

- Single bartender managing security duties simultaneously
- Reactive response to incidents (not preventive)
- Safety risk for staff & patrons

Tertiary: Capacity & foot traffic constraints

- Limited venue size restricts event potential
- Safety concerns deter new customers from attending

- Underutilized capacity due to reputation/safety issues

Willingness to Pay

WTP Range: $250/month ($3000/year)

Drivers:

- Incident prevention (50%): Solve real, frequent problem (bar brawls)
- Liability reduction (30%): Reduce lawsuit/legal exposure risk
- Foot traffic increase (15%): Safety reputation boost attracts new customers
- Staff safety (5%): Protection for bartender & staff

Strong Signal: Owner “very much interested” & sees this as solving a “real problem”

Interview 3 - 230 Fifth Rooftop Bar

Venue: 230 Fifth Rooftop Bar

Interviewee: Floor Manager

Pain Points

Primary: Multi-floor security management & incident prevention

- Frequent events with overcrowding
- Sexual harassment & fights major issues (alcohol-fueled environment)
- Police involvement even when venue not directly responsible
- Multiple floors = distributed security challenge (hard to monitor all areas)
- Current approach: Manual security staff spread thin across floors

Secondary: Capacity constraints due to security limitations

- Operating at 90% capacity during peak time (intentionally reduced)
- Cannot maximize revenue due to lack of adequate security
- Limited waitstaff exacerbates overcrowding issues
- Direct revenue loss from intentional capacity reduction

Tertiary: Liability & legal exposure

- Police involvement creates reputational risk
- Potential negligent security lawsuits despite venue not at fault
- Compliance violations during high-capacity events
- Insurance premium risk from frequent incident reports

Willingness to Pay

WTP Range: The person didn't have much information on the cost and budget and indicated that decision makers will be a lot interested in the solution.

Drivers:

- Incident prevention (40%): Prevent fights, harassment, drunk patron conflicts
- Liability protection (35%): Reduce police involvement & negligent security exposure
- Capacity expansion (20%): Unlock 10% additional capacity = significant revenue upside
- Staff safety (5%): Protect floor managers & security from violent situations

Strong Signal: Recognizes this as critical operational constraint on revenue

Interview 4 - Folly

Venue: Folly Restro-Bar

Location: 92 W Houston St, New York, NY 10012 (Near NYU)

Interviewees: Bartender (Contract), Manager

Pain Points

Primary: Police involvement from theft, fights & harassment

- Frequent incidents (theft, physical altercations, sexual harassment)
- Police called regularly despite venue often not directly responsible
- Liability exposure from negligent security lawsuits
- Reputation damage from police involvement deters customers
- Current approach: Reactive incident response + staff enforcement

Secondary: Weekend overcrowding management

- Heavy foot traffic on weekends strains security capabilities
- Insufficient staff to manage both service & security simultaneously
- Increased incident frequency during peak hours
- Difficulty distinguishing genuine threats in crowded environment

Tertiary: Theft & property crime

- Regular theft incidents (patron belongings, bar inventory)
- Loss of customer trust due to insecurity
- Increased insurance claims

Willingness to Pay

WTP Range: HIGH (exact amount not committed but strong signal)

Drivers:

- Incident prevention (40%): Prevent theft, fights, harassment
- Liability reduction (35%): Reduce police involvement & negligent security exposure
- Customer trust (15%): Safer venue = higher foot traffic & retention
- Brand protection (10%): Reduce reputation damage from crime incidents

Strong Signal: Manager/decision-makers see this as “a challenge” worth investing in; WTP not constrained by budget limitations

Interview 5 - Subway

Interviewee: Store Manager/Franchisee

Pain Points

Primary: Minimal security requirements

- Quick service model = high throughput, low dwell time
- Customers pack food & leave (don’t linger)
- No crowding issues or capacity constraints
- Minimal threat incidents (low-risk environment)

Secondary: No dedicated security infrastructure

- Most QSR outlets rely only on CCTV cameras (no active monitoring)
- No security staff deployed
- Manual staff handles any issues (rare occurrence)

Tertiary: Low operational complexity

- Simple transaction flow (order → payment → pickup)
- Minimal staff interaction beyond food service
- No events, parties, or special crowd management needs

Willingness to Pay

WTP: $0 (NOT INTERESTED / NO WILLINGNESS TO PAY)

Rationale:

- No perceived overcrowding or threat problem
- Current CCTV infrastructure sufficient for operational needs
- Low-risk customer demographic for QSR environment
- VenueShield capabilities = unnecessary overhead

Interview 6 - Arlo Williamsburg Theater

Contact: Rayneet Banerjee
Type: Theater/Performance Venue

Pain Points

Primary: Crowd management during multi-show days

- Frequent performances with audience capacity 500-800
- Ingress/egress congestion during peak showtimes
- Audience safety during packed houses (exits, emergency procedures)
- Current approach: Manual ushers and staff coordination

Secondary: Harassment & disruptive patron incidents

- Occasional confrontations between patrons
- Inappropriate behavior management
- Staff safety concerns during high-volume events

Tertiary: Compliance with venue fire codes & occupancy limits

- Real-time capacity monitoring needed
- Emergency evacuation procedures validation

Willingness to Pay

WTP: $350 - 450/month

Drivers:

Crowd safety management (35%)
Compliance automation (30%)
Staff efficiency (20%)
Brand reputation (15%)

Rationale: Theater venues need proof of compliance; incident prevention protects reputation and ensures customer loyalty for recurring shows.

Interview 7 - The Valencia Room

Contact: Peter Tam
Type: Nightclub

Pain Points

Primary: Frequent bar brawls & aggressive patron behavior

- Weekend incidents (fights, verbal altercations)
- Alcohol-fueled aggression management
- Bouncer-to-patron ratio insufficient during peak hours

Secondary: Overcrowding & safety hazards

- Standing-room-only capacity during peak hours
- Risk of crowd crush or injuries
- Evacuation challenges in packed environment

Tertiary: Drug/substance-related incidents

- Occasional suspicious patron behavior
- Staff concerns about safety with intoxicated guests

Willingness to Pay

WTP: $250-350/month

Drivers:

- Incident prevention (45%)
- Staff safety (25%)
- Liability reduction (20%)
- Insurance premium reduction (10%)

Rationale: Nightclubs face frequent incidents; VenueShield directly prevents costly lawsuits and staff injuries. ROI clear and immediate.

Interview 8 - Icone Club

Contact: Jaqueline Ortiz
Type: Nightclub

Pain Points

Primary: Frequent sexual harassment & assault incidents

- Dance floor harassment common
- Patron boundary violations
- Limited staff ability to monitor all areas simultaneously

Secondary: Unauthorized entry & fake ID issues

- Door staff overwhelmed during busy nights
- Underage drinking concerns

Tertiary: Theft & property crime

- Patron belongings stolen during crowded nights
- Loss of customer trust

Willingness to Pay

WTP: $280 - 380/month

Drivers:

- Safety for female patrons (40%)
- Staff protection (20%)
- Liability reduction (25%)
- Trust & reputation (15%)

Rationale: Modern nightclubs prioritize safety culture; VenueShield enables "safe space" branding, attracting more customers, especially women.

Interview 9 - Desert Lounge

Contact: Diego Natan
Type: Lounge/Bar

Pain Points

Primary: Upscale clientele with high-value items at risk

- Expensive jewelry/handbags frequently targeted for theft
- High-value incident response needed
- Insurance implications for premium clientele

Secondary: VIP patron management & privacy concerns

- Celebrity/high-profile guests requiring discretion
- Need for discreet monitoring without being intrusive
- Conflict de-escalation for premium customers

Tertiary: Compliance with liquor license regulations

- Overserving prevention
- Underage patron prevention

Willingness to Pay

WTP: $220 - 300/month

Drivers:
- Theft prevention (35%)
- VIP safety/discretion (30%)
- Insurance premium reduction (25%)
- Compliance (10%)

Rationale: Lounge model = high-margin per patron; theft/incidents directly reduce profitability. Premium pricing justified by high-value clientele protection.

Interview 10 - Den Social

Contact: Alex Thomas
Type: Nightclub

Pain Points

Primary: Multi-level venue security coordination

- 3 - 4 floors = distributed incident management challenge
- Security blind spots between floors
- Di cult to maintain consistent enforcement across levels

Secondary: Frequent underage drinking violations

- Compliant door checking but enforcement inconsistent
- Risk of liquor license violations

Tertiary: Peak capacity incidents

- Weekend overcrowding
- fire code violations
- Evacuation route management

Willingness to Pay

WTP: $260 - 360/month

Drivers:

- Multi-floor coordination (40%)
- Incident prevention (30%)
- Compliance/fire code (20%)
- Staff efficiency (10%)

Rationale: Multi-floor venues face unique challenges; centralized monitoring justifies premium pricing.

Interview 11 - Desert Lounge Bar

Contact: Sara Khalil
Type: Bar

Pain Points

Primary: Regular bar patron conflicts

- Frequent fights & verbal altercations
- Regular police involvement
- Liability exposure from negligent security

Secondary: Limited security sta ng

- Single bartender often managing security alone
- Cannot monitor entire venue simultaneously
- Response time to incidents too slow

Tertiary: Lost revenue from reputation damage

- Incidents deter new customers
- Operating below capacity due to safety concerns

Willingness to Pay

WTP: $200-280/month

Drivers:

- Incident prevention (40%)
- Reputation management (25%)
- Revenue recovery (25%)
- Staff safety (10%)

Rationale: Bar model = high throughput; one incident can damage reputation for weeks. Justifies investment in prevention.

Interview 12 - La Bloom

Contact: Javier C
Type: Nightclub

Pain Points

Primary: High-volume event management

- Frequent special events & DJ nights (3-4x/week)
- Temporary crowd surges
- Real-time capacity monitoring needed

Secondary: Alcohol-fueled aggression incidents

- Weekend fights common
- Staff injuries from patron altercations

Tertiary: Compliance with local regulations

- Events require special permits
- Ongoing fire code compliance

Willingness to Pay

WTP: $240-340/month

Drivers:

- Event management (30%)
- Incident prevention (35%)
- Compliance automation (20%)
- Staff safety (15%)

Rationale: High-event-frequency venues benefit from automated monitoring. ROI immediate due to frequent incidents.

Interview 13 - Marina Lounge

Contact: Dustin Sullivan
Type: Lounge/Bar

Pain Points

Primary: Upscale bar with sophisticated but aggressive clientele

- High-value patrons with alcohol-related conflicts

- Discreet de-escalation needed without appearing heavy-handed
- Reputation damage from public incidents

Secondary: High theft risk

- Expensive drinks/bar inventory theft
- Patron possession theft

Tertiary: Compliance with premium venue standards

- Liquor license protection Health code compliance

Willingness to Pay

WTP: $210-290/month

Drivers:
- Incident prevention (35%)
- Reputation protection (30%)
- Theft prevention (20%)
- Compliance (15%)

Rationale: Premium positioning = premium willingness to pay for venue protection.

Interview 14 - Temple Night Club

Contact: Ronae Wells
Type: Nightclub

Pain Points

Primary:
- High-capacity nightclub (1000+ capacity) with frequent incidents
- Large venue = distributed security challenge
- Multiple zones of potential conflict
- Significant liability exposure from large incident

Secondary: Frequent overcrowding during weekend events

- Standing-room-only capacity

- Fire code compliance risk
- Evacuation procedure validation

Tertiary: Organized patron groups causing incidents

- Coordinated conflicts between rival groups
- Gang-related activity concerns

Willingness to Pay

WTP: $270-370/month

Drivers:
- Large-scale incident prevention (40%)
- Multi-zone monitoring (25%)
- Liability reduction (20%)
- Fire code compliance (15%)

Rationale: Large venues = large liability exposure. VenueShield ROI measured in single incident prevention.

Interview 15 - Mable's Smokehouse & Bar

Contact: Nicholas Nassour
Type: Bar/Restaurant Hybrid

Pain Points

Primary: Casual bar environment with occasional conflicts

- Infrequent but notable brawls (1-2/month)
- Sports bar atmosphere = occasional aggressive patrons
- Current approach: Manual staff enforcement

Secondary: Happy hour overcrowding

- Brief peak periods (5-7pm) with capacity strain
- Limited temporary staff during peaks

Tertiary: Compliance with food & beverage regulations

- Liquor license protection

Willingness to Pay

WTP: $100-150/month

Drivers:
- Incident prevention (40%)
- Compliance (30%)
- Operational efficiency (30%)

Rationale: Lower incident frequency = lower urgency. WTP driven by risk mitigation, not active prevention. Willing to pay for peace of mind but budget-conscious.

Interview 16 - Bathhouse Williamsburg

Contact: John Gorge
Type: Entertainment/Multi-Purpose Venue

Pain Points

Primary: Diverse event types (DJ nights, live music, events)

- Variable crowd types = different security needs
- Occasional incidents depending on event type
- Manual crowd management during events

Secondary: Limited permanent security staff

- Relies on temporary security for large events
- Inconsistent enforcement across event types

Willingness to Pay

WTP: $130-160/month

Drivers:
- Event-based security (35%)
- Compliance (35%)

- Operational efficiency (30%)

Rationale: Multi-purpose venue = variable needs. WTP moderate because not all events require intensive monitoring.

Interview 17 - Aura Cocina & Bar

Contact: Ethan Xie
Type: Cocktail/Upscale Bar

Pain Points

Primary: Upscale cocktail bar with sophisticated clientele

- Lower incident frequency (mature demographic)
- Occasional high-value patron conflicts
- Premium environment = reputation-sensitive

Secondary: Happy hour compliance

- Overserving prevention during busy periods
- Liquor license protection

Willingness to Pay

WTP: $110-160/month

Drivers:

- Reputation protection (40%)
- Compliance (35%)
- Incident prevention (25%)

Rationale: Upscale positioning attracts mature clientele with lower incident frequency. WTP moderate because primary pain is compliance/reputation, not frequent incidents.

Interview 18 - The Deep Waters

Contact: Johan Pauli
Type: Nightclub

Pain Points

Primary: Lower-intensity nightclub with occasional incidents

- Moderate incident frequency (compared to high-volume clubs)
- Mix of regular patrons with occasional troublemakers
- Manual security su cient but could be optimized

Secondary: Capacity monitoring during events

- Occasional overcrowding concerns
- Fire code compliance

Tertiary: Compliance with venue regulations

Willingness to Pay

WTP: $130-180/month

Drivers:

- Incident prevention (40%)
- Capacity management (30%)
- Compliance (30%)

Rationale: Moderate incident frequency = moderate urgency. WTP reflects need for optimization rather than crisis prevention. Price-sensitive segment.

Interview 19 - Regal Times Square

Contact: Louis P.

Type: Theater

Pain Points

Primary: Theater patron management during peak showtimes

- Mid-to-large capacity theater (600–800 seats)
- Multiple daily performances create ingress/egress congestion

- Theater district foot traffic = mix of tourist and local audiences
- Staff coordination challenges during back-to-back shows
- Current approach: Manual ushers and traditional security procedures

Secondary: Disruptive patron incidents

- Occasional verbal altercations between patrons
- Phone/device disturbances during performances
- Intoxicated patron management (pre-show drinking)
- Discreet de-escalation needed to avoid disrupting shows

Willingness to Pay

WTP: $80-100/month

Drivers:

- Theater-specific crowd management (35%)
- Compliance automation & fire code management (30%)
- Reputation protection (20%)
- Operational efficiency (15%)

Rationale: Theater venues operate on tight margins with high fixed costs. While incident prevention is important, primary need is compliance automation and staff efficiency rather than crisis prevention. Moderate WTP reflects lower incident frequency compared to nightclubs, but higher compliance burden than bars.

Insights and Results

After presenting our demo and thoroughly exploring each venue's pain points, **11 out of 19 stakeholders expressed immediate interest and opted to join our waitlist**. This early conversion rate **approximately 57%** strongly indicates that our solution addresses a timely and meaningful market need.

Such a high willingness to adopt, even prior to launch, provides early validation of both our value proposition and product-market fit. It suggests that with targeted outreach and refined customer segmentation, we can build a strategically scalable and profitable business model.

Additionally, these interviews helped us significantly **de-risk key assumptions** surrounding willingness to pay, urgency of the problem, and overall market readiness. The positive response underscores that our solution resonates with real operational challenges faced by venues today.

Below is a screenshot of the database capturing the details of the venues that have joined our waitlist.

1	Ravneet Banerjee	aybanerj2019@gmail.com	Arlo Williamsburg	Theater	NULL
2	Peter Tam	ptam@valencia.com	The Valencia Room	Nightclub	NULL
3	Jaquline Ortiz	maya@iconsastoria.com	Icons Club	Nightclub	NULL
4	Diego Natan	dnatan@terraz9.com	terraz9	Other	NULL
5	Sara Khalil	sara@desertlounge.co	Desert Lounge Bar	Other	NULL
6	Javier C	javier@laboomqueens.com	La Bloom	Nightclub	NULL
7	Alex Thomas	alex0t@gmail.com	Den Social	Nightclub	NULL
8	Kane Prieto	yagoprieto07@gmail.com	Jimmy's Corner	Other	Frequent bar fights an…
9	Joshua Williams	jwilliams@gmail.com	230 5th Rooftop Bar	Nightclub	NULL
10	Dustin Sullivan	dustinsully@gmail.com	Marina Lounge	Other	NULL
11	Ronae Wells	inquiries@templeclub.c…	Temple Night Club	Nightclub	NULL

Pattern: What Drives High WTP

Interested - Frequent events + High traffic + Overcrowding + Real incidents = $150–300+/month WTP

Not Interested - Low traffic + No events + Minimal threats = $0-100 WTP

Strategic Recommendation

Focus NYC Launch on Event-Driven Venues :

- Professional/Enterprise tier pricing ($250–400+/month)
- High ROI from capacity unlock alone

- Decision-makers actively seeking solutions
- Expected conversion: Very High

Secondary segment: Compliance-focused venues $75–125/month with custom tier

Experiment 2 (Prototype) - Threat-detection accuracy

- **Summary**: This technical component of our project is the core of our product. We want to build a machine learning model that can detect safety hazards in large events and properly classify them. This model has to be able to perform given uncertain lighting conditions, variable crowd densities, and comply with sensitive data management.
- **Hypothesis** (falsifiable): The model can use live camera-feed footage and detect and classify a list of safety hazards with above 90% accuracy AND under 3 second inference latency.
- **Method**:
  - Dataset: streaming video footage from surveillance cameras
  - Model used: YoloV8 - object detection pre-trained computer vision model
  - Duration: 2-3 weeks
- **Variables & Instrumentation.**
  - Independent: detection threshold; class set.
  - Dependent: precision, recall, f1-score
  - Measurement: timestamped logs, event counts, confusion matrices; saved screenshots for errors
- **Success criteria.**
  - Meet or beat precision/recall targets (allow for up to 20% false positives, and <1% false negatives, whilst maintaining 90%+ accuracy and <3 second inference time.)
    - If not met → narrow safety hazard scope (e.g., start with weapons + crowd surge only).
- **Rigor vs. cost.**
  - Can quickly iterate and train in Colab + Google Cloud Credits (<50$)
- **Decision & next step.**
  - Pass: Try gathering real footage to experiment with lighting conditions, etc.
  - Fail: Do more research on possible computer vision models to provide better accuracies

Experiment 3 - Multi-camera ingest & edge latency (prototype, primary)

- **Summary**: This experiment tests the system's ability to ingest footage from diverse camera sources (H8) and process it in real-time on low-power hardware (H10). This is a crucial technical feasibility gate for running a cost-effective pilot on a venue's existing infrastructure.
- **Hypothesis (falsifiable):**
  - Median end-to-end latency (frame → alert) ≤ 1.0s at 720p/10 fps; and system achieves stable ingest (frame drop rate ≤ 1%) from ≥ 2 diverse camera sources for continuous 30 minute run
- **Method**:
  - **Sources**: Ingest and process simultaneous streams from three distinct camera types to simulate real-world variability:
    - (1) Standard IP cam stream (e.g., Hikvision/Axis)
    - (2) Low-resolution/legacy analog feed (via encoder)
    - (3) Mobile phone RTSP feed (as a proxy for generic software source).
  - **Edge Device**: Utilize a low-cost, low-power edge compute device (e.g., Jetson Nano/RPi 5) to run the fixed object detection model from Experiment 1.
  - **Duration**: 3–4 days (setup + data collection).
  - **Procedure**: Run the ingestion-processing pipeline for ≥ 30 continuous minutes per camera type, measuring E2E latency and frame drop rate simultaneously.
- **Variables & Instrumentation.**
  - **Independent**: Camera Type/Bitrate; Edge-Only vs. Edge+Cloud processing path (if applicable)
  - **Dependent**: Median and 95th percentile End-to-End Latency, Frame Drop Rate (per camera source), CPU/GPU Utilization.
  - **Controls:** Fixed object detection model/settings (from Exp 1), Fixed resolution (720p) and FPS (10 fps).
  - **Measurement**: Primary data from timestamped logs for: frame ingestion, alert generation, and dropped frames to compute latency and stability. OS metrics for CPU/GPU load.
- **Success Criteria**.
  - **Pass**:
    - (1) Median E2E Latency ≤ 1.0s
    - (2) Stable ingest (Frame Drop Rate ≤ 1%) on ≥ 2 of the 3 tested camera types.
- **Rigor vs. Cost**.
  - High rigor, low cost. Hardware cost ≤ $150 plus engineering time.

    - This test provides decisive primary behavioral data on system performance under the high uncertainty of diverse camera hardware (H8) and the high importance of real-time performance (H10).
    - It is a critical, cheap gate before any field deployment.
- **Decision & Next Step.**
    - **Pass**: Proceed to Experiment 3 (Resilience Simulation) and confirm that the pilot can rely on existing venue camera hardware.
    - **Fail:** Pivot by:
        - (1) Updating the pilot specifications to only support a narrower set of camera brands
        - (2) Optimizing the ML pipeline (e.g., model compression, reduced resolution) and re-running the test.

Experiment 4 - Resilience under packet loss/overload (prototype, sim)

- Summary:
- Hypothesis (falsifiable).
    - If we demonstrate resilience via prototype simulation, organizers' perceived trust score increases over 30%, purchase intent rises over 20%.
- Method:
    - Inject loss/throttle via network tools while streaming two sources for 45 minutes.
    - Duration: 2–3 days.
- Variables & Instrumentation.
    - Independent: loss %; throttle level; single vs. dual stream.
    - Dependent: availability; recovery time; error rate; alert backlog.
    - Controls: fixed thresholds; same clips as Exp 2.
    - Measurement: uptime logs; error dashboards; recovery timestamps.
- Success criteria.
    - Availability ≥ 99%; median recovery ≤ 5 s; zero fatal crashes.
- Rigor vs. cost.
    - Very low cost; high confidence gain before fielding
- Decision & next step.
    - Pass: finalize pilot SLO/SLA + incident runbook.
    - Fail: add retry/back-pressure; defer pilot until stability proven.